INTERNATIONAL TELECOMMUNICATION UNION

TELECOMMUNICATION STANDARDIZATION SECTOR

STUDY PERIOD 2025-2028

FOCUS GROUP ON AI NATIVE FOR TELECOMMUNICATION NETWORKS (FG-AINN)

FG-AINN-I-xxx

Original: English

Question(s): N/A

New Delhi, 09th October, 2025

INPUT DOCUMENT

Source:	Manipal Institute of Technology, Bengaluru	
Title:	Build-a-thon 3.0 2025: Autonomous Energy Aware Self Optimizing Radio Access Network (Auto EA-SORAN)	
Contact:	Anirudh Nishtala Dept. of ECE, 4th Year Manipal Institute of Technology, Bengaluru	E-mail: anirudh2.mitblr2022@learner.manipal.edu
Contact:	Rishiraj Rakesh Kumar Dept. of ECE, 4th Year Manipal Institute of Technology, Bengaluru	E-mail: rishiraj.mitblr2022@learner.manipal.edu
Contact:	Tarunika D Dept. of ECE, 4th Year Manipal Institute of Technology, Bengaluru	E-mail: tarunika.mitblr2022@learner.manipal.edu
Contact:	Sreelakshmi Anil Dept. of ECE, 4th Year Manipal Institute of Technology, Bengaluru	E-mail: sreelakshmi.mitblr2022@learner.manipal.edu
Contact:	Dr. Shweta Singh Asst. Prof., Dept of ECE Manipal Institute of Technology	E-mail: shweta.s@manipal.edu
Contact:	Mr. Rajat Duggal Senior System Specification Engineer MN RAN R&D, Nokia Bengaluru	E-mail: rajat.duggal@nokia.com
Contact:	Mr. K. Kanagasundaram Principal Technical Specialist NI IPN NW A NSP R&D India 7 (L06), Nokia Bengaluru	E-mail: kanagasundaram.k@nokia.com

Abstract: This document describes a submission towards Build-a-thon 3.0, 2025 conducted in New Delhi on October 9th, 2025.

1. Introduction

1.a) Background to the Topic: The Energy Consumption Imperative

While the 5G New Radio (NR) standard is significantly more energy-efficient per bit than 4G LTE, the total energy consumption of 5G networks is projected to rise dramatically. This paradox is driven by network densification, massive Multiple-Input Multiple-Output

(MIMO) antenna arrays, and exponential growth in data traffic required to meet 5G's performance goals.

These factors create significant operational and environmental challenges. The Radio Access Network (RAN) accounts for 75-85% of a mobile network's energy use, making it a major operational expenditure (OPEX) that impacts financial sustainability. Operators also face pressure to meet sustainability goals like the UN's SDG-13. A key inefficiency is the "always-on" nature of base stations, which consume high power even during low traffic periods, indicating a mismatch between static resource provisioning and dynamic demand.

1.b) Need for AI-Native Network Automation

The complexity of 5G RANs makes traditional, rule-based Network Management Systems (NMS) obsolete, as they are too static, manual, and reactive for 5G's dynamic environment. The necessary shift is towards AI-native networks that embed intelligence into the network fabric, enabling autonomous, proactive control through distributed, agent-based architectures. This creates a fully automated closed loop of observation, prediction, decision, and actuation.

AI-native automation is also essential for 5G's core business case of network slicing, which supports diverse services like enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), and massive Machine-Type Communications (mMTC) on a single infrastructure. These services have conflicting energy and performance needs. A static energy-saving policy would either violate Service Level Agreements (SLAs) or miss optimization opportunities. Therefore, an intelligent system that dynamically manages the energy-QoS trade-off on a per-service basis is a foundational requirement.

2. State of the Art in 5G Energy Management

2.a) Commercial AI-Powered Platforms

Major telecom vendors have integrated AI-driven energy management into their 5G RAN portfolios, moving from static configurations to dynamic, data-driven optimization.

- One vendor uses an ML-based engine to set dynamic thresholds for cell sleep modes, reporting energy reductions of 10-12%.
- Another uses MantaRay Self-Organizing Network (SON) platform enables closed-loop automation, claiming up to 30% additional power reduction.
- A third uses AI-based traffic prediction to automate features like power amplifier control and multi-level radio sleep modes.

While these platforms show a trend towards intelligent automation, they are often proprietary systems that may lack the transparency of more advanced frameworks.

2.b) Academic Research Contributions

Academic research has applied machine learning to 5G energy efficiency across several key areas. Foundational work includes **traffic prediction and load forecasting** using models like LSTMs to enable proactive control. Building on this, a dominant theme is **dynamic base station (BS) on/off switching**, using techniques from heuristics to Reinforcement Learning (RL) to manage sleep modes. Other significant areas include the use of **Multi-Agent Systems (MAS)** for robust, distributed control ; ML-based optimization of **resource allocation** ; and **Federated Learning (FL)** for privacy-preserving model training.

2.c) Standardisation Efforts in Energy Efficiency

Standardization work by 3GPP's Working Group 5 (SA5) provides the foundation for scalable network automation, defining standards for Energy Efficiency (EE) and Energy Saving (ES) across the entire 5G system. A key contribution is the formal definition of the EE KPI:

$$EE = \frac{Performace}{Energy\ Consupmtion\ (EC)}$$

This ratio connects energy use to network output, with "Performance" defined differently based on service type (e.g., data volume for eMBB, reliability for URLLC). 3GPP also specifies standardized methods for data collection via OA&M interfaces. This creates a vendor-neutral language for EE, providing a credible and interoperable target for AI optimization algorithms and making the use of 3GPP-defined KPIs a critical design choice.

3. Research Gaps and System Limitations

Analysis of the state of the art reveals key gaps that the Auto EA-SORAN project addresses.

- **Holistic vs. Siloed Designs:** Many solutions optimize single parameters in isolation, such as cell sleep modes. This ignores interdependencies between control levers. The proposed multi-agent architecture enables holistic optimization by coordinating multiple actions simultaneously, such as transmit power and sleep schedules, to find superior solutions.
- **The High-Fidelity Validation Gap:** Much academic research uses simplified simulators that miss complex, real-world protocol stack effects, like how power reduction can increase MAC layer retransmissions. This project uses Simu5G as a high-fidelity digital twin that models the full 5G stack, ensuring control policies are robust. This also solves the "data gap" by generating realistic synthetic data for model training.
- **The Explainability Deficit:** The "black box" nature of many AI models creates a trust deficit for operators. This project builds in *architectural explainability*. The **Feedback Agent** is not a complex model but a clear logic gate that validates every action against understandable, 3GPP-compliant performance thresholds, making the system's behaviour transparent by design.
- **Static Management of the QoS-Energy Trade-off:** Current systems often use static, network-wide thresholds to manage the trade-off between energy savings and QoS. This is ill-suited for 5G's diverse services. The proposed system manages this trade-off dynamically with service-level granularity, using the Feedback Agent to apply different validation policies for different traffic types, thus maximizing savings without violating SLAs.

4. Technological Foundations and Proposed Architecture (Auto EA-SORAN)

4.a) A Closed-Loop Multi-Agent Control System

The Auto EA-SORAN system is a closed-loop control architecture composed of four cooperative agents executing a continuous cycle of monitoring, prediction, optimization, and actuation.

- **Prediction Agent:** Ingests real-time network state data and uses an ML model to forecast future energy consumption.
- **Optimization Agent:** Receives the forecast and formulates a multi-faceted energy conservation plan (e.g., adjust power, enter sleep mode).
- **Feedback Agent:** Validates the proposed plan against predefined policies and 3GPP-compliant performance thresholds to prevent QoS degradation.
- **Orchestration Agent:** Executes the final, validated plan within the simulation environment.

4.b) Simu5G as a High-Fidelity Digital Twin

The system is developed and validated within the Simu5G network simulator, which acts as a high-fidelity digital twin of the 5G RAN, modeling the complete NR protocol stack. A key innovation is a custom, state-based gNB energy model built by extending INET's Energy Framework. It defines discrete power states (e.g., active, micro-sleep, deep sleep) and models active power consumption as a dynamic, linear function of real-time Physical Resource Block (PRB) utilization:

$$Ptotal = Pbase + Ptraffic \times \left(\frac{num_allocated_PRBs}{total_available_PRBs}\right)$$

This ensures the model is realistically coupled with data plane activity. To maintain a responsive control loop, the architecture uses a low-latency, socket-based Inter-Process Communication (IPC) mechanism (ZeroMQ), between the Python agents and the C++ simulation.

5. Current Status & Demo Proposal

Our project, “Autonomous Energy Aware Self Optimizing RAN,” will culminate in a demonstration of a complete, closed-loop optimization cycle within the simulated 5G network. The demonstration will visualize the following sequence:

- The system observes the network state in Simu5G (e.g., PRB utilization, active user count).
- The Prediction Agent forecasts future traffic and energy consumption.
- The Optimization Agent proposes a specific energy-saving action (e.g., "transition gNB-3 into state_micro_sleep").
- The Feedback Agent validates that this action will not violate the predefined QoS policies for active users.
- The Orchestration Agent successfully transmits and executes the command within the Simu5G environment.
- The results will be shown via the simulation's output vectors and scalars, illustrating a measurable decrease in the target gNB's power draw while maintaining stable user throughput. A comparative analysis against a baseline (unoptimized) simulation run will be presented to quantitatively demonstrate the percentage of energy saved.

Sub missi on Id	FG-AINN-PoC-xxx (Id number is to be assigned by the secretariat)
PoC Obje	In this Build-a-thon proposal, we aim to study **AI-native, closed-loop energy optimization for 5G base stations** using a prediction → optimization → feedback→ actuation cycle inside **Simu5G**.

ctive	**Specifically,** we will demonstrate: 1. a **Prediction Agent** that forecasts short-horizon traffic and energy, 2. an **Optimization Agent** that selects an energy-saving action (e.g., TX-power trim, carrier off, micro/deep sleep) under **QoS guard-rails**, 3. a **Feedback/Safety Agent** that validates and can **roll back** actions, and 4. an **Orchestration Agent** that executes the command in Simu5G via a low-latency socket. **At this point, we hypothesize that** this closed loop will deliver **≥15–25% energy reduction** on targeted gNBs **without violating QoS** (≤3% p95 throughput degradation; latency/loss within policy).
PoC Setup	In this proposal we use the **Farzi project's three CSV datasets** (static, offline) **plus real-time telemetry from Simu5G**. The three CSV files train the energy-prediction model; **Simu5G** feeds the trained model with live network metrics (e.g., PRB utilization, active UEs, throughput/latency) during the demo. **Model training is done using** Python (**scikit-learn/Keras**) with engineered temporal features; **SHAP** provides model-level explainability. The agents communicate with an in-simulation **Orchestration Agent** over **ZeroMQ/TCP**; energy is measured via a **state-based gNB consumer** (active / micro-sleep / deep-sleep) that depends on PRB utilization.
PoC demo and Success Criteria	The following scenarios would be demonstrated: 1) **Closed-loop saving:** the system predicts a low-load window and executes an action (e.g., "gNB-3 → micro_sleep" or "ΔP = −2 dB"); vectors/scalars show **lower power** while UE throughput remains within policy. 2) **Intent-to-policy orchestration:** an operator intent such as "**minimize energy with p95 UE throughput ≥ X and latency ≤ Y**" is translated to policies; when conditions are met, an energy action is applied and logged with an **explanation trace**. 3) **Safety & rollback:** if post-actuation KPIs breach guard-rails, the **Feedback Agent** auto-reverts to the last safe configuration and records the event.
PoC Plan and Report	Stage-1: Demo proposal by September 07, 2025 Stage-2: Implementation proposal and test cases by September 15, 2025 Stage-3: Initial implementation by September 30, 2025 Stage-4: Final demo and report in FG AINN template by October 9, 2025
Created by	Team: NoWiresAttached Team lead: Ms. Tarunika D. Team members: 1.Mr. Anirudh Nishtala 2.Mr. Rishiraj Rakesh Kumar 3.Ms. Sreelakshmi Anil Internal guide: Asst. Prof. Dr. Shweta Singh
Creation date	05 September 2025
Ope	While our current proof-of-concept demonstrates prediction and autonomous self-optimization

n problems and future study	through explainable AI and lightweight agents, several open challenges remain for further research and deployment at scale: 1. **Multi-cell coordination:** energy savings with HO/load-balancing awareness across clusters. 2. **Real-time constraints:** tighter inference/optimization deadlines and jitter control on commodity edge hardware. 3. **Policy richness:** extend guard-rails to include SLA tiers, slice-specific constraints, and time-of-day pricing signals. 4. **Model drift & continual learning:** handle seasonality and traffic shifts; evaluate **federated**/online learning. 5. **Wider actuator set:** PA bias control, beam/PRB budgeting, scheduler knobs; safety proofs for each. 6. **Standards alignment:** map to **O-RAN** E2 control (near-RT RIC xApps) for field portability. 7. **Adversarial robustness:** spoofed telemetry/conflicting intents; formal verification of rollback logic. These open problems provide an opportunity for iterative enhancement of our framework toward a production-ready, standard-compliant AI-Native self-optimizing system suitable for the energy conservation of 5G Base Stations.
References	[1] "A Calibrated and Automated Simulator for Innovations in 5G," *arXiv*, 2024. https://arxiv.org/html/2404.10643v1 [2] "Scalable Real-Time Emulation of 5G Networks With Simu5G," *ResearchGate*, n.d. https://www.researchgate.net/publication/355690617 [3] "Detailed Explanation on 5G Simulation: Parameters • Model • Indicators," *OMNeT++ Tutorial*, n.d. https://omnet-tutorial.com/5g-simulation/ [4] "Scalable Real-Time Emulation of 5G Networks With Simu5G," *SciSpace (PDF)*, n.d. https://scispace.com/pdf/scalable-real-time-emulation-of-5g-networks-with-simu5g-2mv2rbff4t.pdf [5] "5G-Energy-Consumption-Modelling-Solution — Team Farzi Data Scientists," *GitHub (ITU AI/ML in 5G Challenge)*, n.d. https://github.com/ITU-AI-ML-in-5G-Challenge/5G-Energy-Consumption-Modelling-Solution-Team-Farzi-Data-Scientists [6] "How to Implement 5G Network Simulator Projects (Guide)," *Network Simulation Tools*, n.d. https://networksimulationtools.com/5g-network-simulator/ [7] "A Survey on Green Enablers: A Study on the Energy Efficiency of AI-Based 5G Networks," *MDPI (Sensors)*, 2024. https://www.mdpi.com/1424-8220/24/14/4609 [8] "Overview — Simu5G: Simulator for 5G New Radio Networks," *Simu5G Users' Guide*, n.d. https://simu5g.org/users-guide/overview [9] "Running Simu5G as a Real-Time Network Emulator," *Simu5G Users' Guide*, n.d.

https://simu5g.org/users-guide/emulation

[10] "Simu5G: A System-level Simulator for 5G Networks," *SciTePress (PDF)*, 2020. https://www.scitepress.org/Papers/2020/98264/98264.pdf

[11] "Simu5G: Simulator for 5G New Radio Networks," *Project site*, n.d. https://simu5g.org/

[12] "Modeling Power Consumption," *INET 4.5.4 Documentation*, n.d. https://inet.omnetpp.org/docs/users-guide/ch-power.html

[13] L. M. Feeney, "An Extensible Framework for Simulating Battery Consumption in Wireless Networks," n.d. https://sonri.org/publications/Files/omnet10energy.pdf

[14] "How to Calculate Network Energy Utilization in OMNeT++," *OMNeT++ Manual (community)*, n.d. https://omnet-manual.com/how-to-calculate-network-energy-utilization-in-omnet/

[15] "AI-based Energy Consumption Modeling of 5G Base Stations: An Energy-Efficient Approach," *figshare (conference contribution)*, n.d. https://sussex.figshare.com/articles/conference_contribution/AI-based_energy_consumption_modeling_of_5G_base_stations_an_energy_efficient_approach/26394829

[16] "Modeling of IoT Devices Energy Consumption in 5G Networks," *mediaTUM (PDF)*, n.d. https://mediatum.ub.tum.de/doc/1705687/55ahjo0udp2jr6z1159ilkwbr.Modelling_energy_consumption_of_IoT_devices_in_5G_network.pdf

[17] "Energy Efficiency for 5G and Beyond 5G: Potential, Limitations, and Future Directions," *MDPI (Sensors)*, 2024. https://www.mdpi.com/1424-8220/24/22/7402

[18] "Energy Consumption Modelling for 5G Radio Base Stations with Machine Learning," *Lund University Publications (PDF)*, n.d. https://lup.lub.lu.se/student-papers/record/9133984/file/9133985.pdf

[19] "BaseStationStatsCollector," *Simu5G NED Documentation*, n.d. https://simu5g.org/neddoc/simu5g.corenetwork.statsCollector.BaseStationStatsCollector.html

[20] "Packages — Simu5G," *Simu5G NED Documentation*, n.d. https://simu5g.org/neddoc/packages.html

[21] "OMNeT++ 5G Thesis Ideas," *PHD Services*, n.d. https://phdservices.org/omnet-5g/

[22] "New Radio Tutorial — Simu5G," *Tutorial docs*, n.d. https://simu5g.org/tutorials/NR/doc/index

[23] "BackgroundScheduler," *Simu5G NED Documentation*, n.d. https://simu5g.org/neddoc/simu5g.nodes.backgroundCell.BackgroundScheduler.html

[24] “Simu5G — 5G NR and LTE/LTE-A User-Plane Simulation Model for OMNeT++ & INET,” *GitHub*, n.d.
https://github.com/Unipisa/Simu5G

[25] “A Network Simulator-based Method of Dynamic Traffic Generation for 5G Network Slicing,” *Concordia University — Spectrum*, n.d.
https://spectrum.library.concordia.ca/id/eprint/990258/1/Ding_MCompSc_df

[26] “Parameter Studies,” *OMNeT++ Technical Articles — TicToc Tutorial*, n.d.
https://docs.omnetpp.org/tutorials/tictoc/part7/

[27] “gNodeB,” *Simu5G NED Documentation*, n.d.
https://simu5g.org/neddoc/simu5g.nodes.NR.gNodeB.html

[28] “OMNeT++ — Simulation Manual,” *Omnetpp.org*, 2025.
https://doc.omnetpp.org/omnetpp5/manual/

[29] “OMNeT++ Simulation Library: cSimpleModule Class Reference,” *Omnetpp.org (API v4)*, n.d.
https://doc.omnetpp.org/omnetpp4/api/classcSimpleModule.html

[30] “Simu5G: A System-level Simulator for 5G Networks,” *ARPI (PDF)*,
https://arpi.unipi.it/retrieve/handle/11568/1039566/609208/SIMULTECH_2020.pdf

[31] “Simu5G – An OMNeT++ Library for End-to-End Performance Evaluation of 5G Networks,” *ARPI (PDF)*, 2020.
https://arpi.unipi.it/bitstream/11568/1053174/2/2020%20Access.pdf

[32] “Steps to Implement Network Energy Management in OMNeT++,” *OMNeT++ Manual (community)*, n.d.
https://omnet-manual.com/how-to-implement-network-energy-management-in-omnet/

[33] “Energy Efficiency (EE) SA5 Work and Results,” *3GPP*, n.d.
https://www.3gpp.org/technologies/energy-efficiency-ee-sa5-work-and-results

[34] “5G and Energy Efficiency,” *5G-PPP Standards Tracker (booklet)*, n.d.
https://standards-tracker.5g-ppp.eu/sites/default/files/BookletA4_EnergyEfficiency.pdf

[35] “A Comparison Analysis of 5G Key Performance Indicators Based on Entropy,” *ResearchGate (PDF)*, n.d.
https://www.researchgate.net/publication/386292921

[36] “Main Modules of the Simu5G Model Library (figure),” *ResearchGate — Scientific Diagram*, n.d.
https://www.researchgate.net/figure/Main-modules-of-the-Simu5G-model-library_fig3344434106

[37] “SDAP-based QoS Flow Multiplexing Support in Simu5G for 5G NR Simulation,” *arXiv*, 2025.
https://arxiv.org/html/2508.12785v1